

POWERFACTORY

PowerFactory 2024

Technical Reference

Connection Request

ElmConreq

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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January 19, 2024
PowerFactory 2024
Revision 2

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1 General Description

For power quality assessment, *PowerFactory* offers a Connection Request Assessment command (*ComConreq*) and corresponding element (*ElmConreq*). The connection request element (*ElmConreq*) is used to model planned connections of generating stations and/or loads for new connectees. The Connection Request Assessment command (described in the User Manual), in conjunction with the Connection Request element, facilitates the execution of a power quality assessment according to the *Method* selected in the command.

The following points are important to note:

- The setting in the command determines the dialog page (*ElmConreq*) from which the infeed will be taken for other calculations such as load flow, short-circuit, etc.; or it will use the first ticked option in the *ElmConreq*.
- All output window messages issued should be noted.
- The connection request element is *only* intended for use in conjunction with the Connection Request command and the calculations that it carries out; i.e. balanced load flow and short-circuit according to DIN EN 60909-0 (VDE 0102).

2 D-A-CH-CZ 2nd Edition

The selection of *D-A-CH-CZ* as the *Method* in the Connection Request Assessment command carries out a power quality assessment according to [1]. The associated input data for the connection request element is described in this section.

2.1 Installation

2.1.1 Loads

- ***Connected load, Sa in kVA.***

2.1.2 Generating units

- ***SrEtot.*** Specifies which value to use in the calculation of SrEtot. If *N/A* is selected, a value of 1 will be used.
- ***cos(phiA).*** The power factor of the generating station.

Table ***Generating units*** per row:

- ***Name.*** An arbitrary name used to describe the generating unit/s.
- ***Type.*** The type of generating unit.
- ***Qty.*** The number of generating unit/s.
- ***Rated voltage, Ur*** The rated voltage of the generating unit/s in kV.
- ***SrE.*** The rated apparent power of the generating unit/se.
- ***cos(phiE).*** The power factor of the generating unit/s.
- ***PnG.*** The rated active power of the generating unit/s.
- ***p0.2.*** Instantaneous value in p.u..
- ***p60.*** 1-minute mean value in p.u..
- ***p600.*** 10-minute mean value in p.u..

2.2 Connection topology

- ***Junction point definition*** Automatic search or user-defined selection of the junction point (PCC).
- ***Automatic detection of supply transformers.*** If ticked, *PowerFactory* will search for all supply transformers using the criterion that they are connected between the voltage level of the PCC and a higher voltage level. If this option is left unticked, the user may specify the supply transformers in the table provided. Relevant to HV networks only.
- ***Supply transformers.*** User-specified supply transformers. These must be connected between the same voltage level as the PCC and a higher voltage level. Relevant to HV networks only.

- ***SNHS.*** Assumed (or indicative) network power of the HV network. This is the sum of the rated power of all supply transformers. Relevant to HV networks only.

2.3 Voltage changes and flicker

2.3.1 Loads

- ***Input mode.*** Selection of mode of data input.
- ***Apparent power change, dSa.*** Apparent power change in kVA for devices and installations, relevant to the assessment of network disturbances.
- ***Angle, phi.*** The angle of the installation of the network user in degrees.
- ***Displacement factor, cos(phi).*** The quotient of the active power and the apparent power, relative to the fundamental voltage and current.
- ***Active power change, dPa.*** Active power change in kW for devices and installations, relevant to the assessment of network disturbances.
- ***Reactive power change, dQa.*** Reactive power change in kvar for devices and installations, relevant to the assessment of network disturbances.
- ***Rated voltage, Ur.*** The rated voltage of the motor in V. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Rated current, Ir.*** The rated current of the motor in A. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Starting current ratio, Ia/Ir.*** The ratio of the motor starting current to the motor rated current, both in A. Valid range is [3-8]. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Load technology.*** The type of connection of the user installation (3PH, 1PH PH-PH, 1PH).
- ***Transformer characteristic number.*** The transformer characteristic number.
- ***Repetition rate.*** The repetition rate in 1/min. *PowerFactory* uses the Pst=1 curve (see [1]) for the calculation of flicker, and therefore supports repetition rates $\leq$ 1800/min.
- ***Shape factor.*** The shape factor. Used for the conversion of special shapes of voltage change characteristics into flicker-equivalent voltage jumps. The shape factor is considered in the calculation of the short- and long-term flicker severity, and in the calculation of the flicker impression time.

2.3.2 Generating units

- ***SrEtot.*** Specify which value to use in the calculation of SrEtot. If *N/A* is selected, a value of 1 will be used.
- ***SrEmax.*** Specify which value to use in the calculation of SrEmax. If *N/A* is selected, a value of 1 will be used.

Table ***Generating units*** per row:

- ***Handle as.*** For the purpose of voltage change and flicker calculations, specifies whether the generating unit/s should be handled as wind turbine generators (WTGs).
- ***Flk. Coeff. (TypFlicker).*** Selection of *Flicker* type in which to specify flicker data for the generating unit/s. This data should only include maximum values.
- ***Ia in kA.*** The rated current of the generating unit/s.
- ***phi_f in deg.*** Flicker-relevant angle of the generating unit/s.
- ***phi_a in deg.*** Starting angle of the generating unit/s.
- ***r in 1/min.*** Repetition rate of the generating unit/s.

2.4 Voltage unbalance

2.4.1 Loads

- ***Load technology.*** The type of connection of the user installation (3PH, 1PH PH-PH, 1PH).
- ***Pre-existing voltage unbalance level in entire grid.*** The voltage unbalance level in % pre-existing in the network under assessment. Relevant to HV networks only.
- ***Pre-existing two-phase installations in entire grid.*** Should be ticked if there are pre-existing two-phase installations in the network under assessment. Relevant to HV networks only.

2.4.2 Generating units

- ***Technology.*** The type of connection (3PH or 1PH) of the generating unit/s in this row of the table.
- ***Phase.*** The phase to which the generating unit/s are connected.

2.5 Harmonics

This input data applies to both loads and generating units. It should be noted that Harmonics will only be assessed for generating units whose *Type* on the *Installation* subpage is set to *Generator with converter*.

- ***Input of device data: According to groups.*** Device data used for calculation of the harmonic load. Input according to Groups defined in [1] and [4].
- ***Input of device Data: According to total harmonic distortion (THDi).*** Device data used for calculation of the harmonic load. Input according to total harmonic distortion is automatically classified into corresponding Groups (as defined in [1] and [4]) and displayed as read-only data in the table.
- ***Type.*** Device type (load or generating unit/s).
- ***Maximum connectible power of all consumer installations.*** Used in the calculation of harmonic current emission limits and subsequently, THDiA. Relevant to HV networks only.

- ***Simultaneity factor.*** Used in the calculation of harmonic current emission limits and subsequently, THDiA. Relevant to HV networks only.
- ***Network level factor.*** Used in the calculation of allowable harmonic voltages, and subsequently, for the calculation of *general* harmonic current emission limits and THDiA. Relevant to HV networks only.
- ***System capacitance.*** Used in the calculation of the first parallel resonance point. Input must already consider length. If not specified, and option *calculated* is selected, *PowerFactory* will calculate the capacitance considering all lines/cables and purely capacitive shunts connected at the same voltage level as the PCC. Relevant to HV networks only.

2.6 Commutation notches

This input data applies to both loads and generating units.

- ***Converter rated power.*** Rated power of the converter in kVA. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Number of pulses.*** Number of pulses of converter. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Worst-case converter angle.*** Worst-case angle of the converter in degrees. Used in the calculation of the commutation reactance short-circuit voltage. Relevant to HV networks only.

2.7 Interharmonic voltages

- ***Input of interharmonics for entire installation: Harmonic voltages (spectral lines.)*** Table of interharmonic voltage magnitudes, defined according to spectral lines. Harmonic frequencies must be entered in ascending order.
- ***Input of interharmonics for entire installation: Harmonic current injections (spectral lines).*** Table of interharmonic current injections, defined according to spectral lines. Harmonic frequencies must be entered in ascending order.
- ***Input of interharmonics for entire installation: Harmonic voltages (groups).*** Table of interharmonic voltage magnitudes, defined according to groups.
- ***Input of interharmonics for entire installation: Harmonic current injections (groups).*** Table of interharmonic current injections, defined according to groups.
- ***Devices (HV Networks only).*** List of devices which create interharmonics. The harmonic load content is calculated as either: the sum of the nominal powers of all identical units; or, the nominal power of the highest-rated (power) unit. Relevant to HV networks only.
- ***Converter power (Devices and installations).*** Used to calculate the ratio of short-circuit power to converter power. Relevant to LV and MV networks only.
- ***Ripple control frequency.*** The frequency of the ripple control in Hz.
- ***Converter type: frequency converter.***
- ***Converter type: Pulse-width modulated rectifier.***

3 D-A-CH-CZ 3rd Edition

The selection of *D-A-CH-CZ* as the *Method* in the Connection Request Assessment command carries out a power quality assessment according to [8]. The associated input data for the connection request element is described in this section.

3.1 Installation

3.1.1 Loads

Table ***Loads*** per row:

- ***Name.*** An arbitrary name used to describe the load/s.
- ***Type.*** The type of load.
- ***Qty.*** The number of load/s.
- ***Rated voltage, Ur*** The rated voltage of the load/s in kV.
- ***SrB.*** The rated apparent power of the load/s.
- ***cos(phiB).*** The power factor of the load/s.

3.1.2 Generating units

- ***SrEtot.*** Specifies which value to use in the calculation of SrEtot. If *N/A* is selected, a value of 1 will be used.
- ***cos(phiA).*** The power factor of the generating station.

Table ***Generating units*** per row:

- ***Name.*** An arbitrary name used to describe the generating unit/s.
- ***Type.*** The type of generating unit.
- ***Qty.*** The number of generating unit/s.
- ***Rated voltage, Ur*** The rated voltage of the generating unit/s in kV.
- ***SrE.*** The rated apparent power of the generating unit/s.
- ***cos(phiE).*** The power factor of the generating unit/s.
- ***PnG.*** The rated active power of the generating unit/s.
- ***p0.2.*** Instantaneous value in p.u..
- ***p60.*** 1-minute mean value in p.u..
- ***p600.*** 10-minute mean value in p.u..

3.1.3 Loads and generating units

General quantities used for both loads and generating units.

- ***SA.*** Apparent connection power of the installation.
- ***kB.*** Expansion factor for load units.
- ***kE.*** Expansion factor for generation units.
- ***kS.*** Expansion factor for storage units.

3.2 Connection topology

- ***Junction point definition*** Automatic search or user-defined selection of the junction point (PCC).
- ***Automatic detection of supply transformers.*** If ticked, *PowerFactory* will search for all supply transformers using the criterion that they are connected between the voltage level of the PCC and a higher voltage level. If this option is left unticked, the user may specify the supply transformers in the table provided.
- ***Supply transformers.*** User-specified supply transformers. These must be connected between the same voltage level as the PCC and a higher voltage level.
- ***SrT.*** Total rated transformer power. This is the sum of the rated power of all supply transformers.

3.3 Voltage changes and flicker

- ***Emission limits calculation.*** Selection of mode for calculation of emission limits either ***Individually per device*** or ***Plant load change***.

3.3.1 Individually per device

3.3.1.1 Loads and generating units

General quantities used for both loads and generating units.

- ***Input mode.*** Selection of mode of data input.
- ***Transformer characteristic number.*** The transformer characteristic number.
- ***Summation exponent, alpha.*** The summation exponent α. The summation exponent is considered in the calculation of the resulting long-term flicker severity of multiple units or plants.
- ***Simultaneity factor, g.*** The simultaneity factor for flicker generating units/plants at the same line section.

3.3.1.2 Loads

Table ***Loads*** per row:

- ***Technology.*** The type of connection of the load/s installation (3PH, 1PH PH-PH, 1PH).
- ***Apparent power change, dSa.*** Apparent power change in kVA for load/s, relevant to the assessment of network disturbances.
- ***Angle, phi.*** The angle of load/s in degrees.
- ***Displacement factor, cos(phi).*** The quotient of the active power and the apparent power of the load/s, relative to the fundamental voltage and current.
- ***Active power change, dPa.*** Active power change of load/s in kW, relevant to the assessment of network disturbances.
- ***Reactive power change, dQa.*** Reactive power change of load/s in kvar, relevant to the assessment of network disturbances.
- ***Impedance ratio, alpha (Zn/Zp).*** Ratio of the impedance of the neutral wire (Zn) to the outer conductor (Zp) of load/s. Used for calculation of the relative voltage change at the operating point of the load/s.
- ***Repetition rate, r.*** The repetition rate of load/s in 1/min. *PowerFactory* uses the Pst=1 curve (see [8]) for the calculation of flicker, and therefore supports repetition rates $\leq$ 12000/min.
- ***Shape factor, F.*** The shape factor of load/s. Used for the conversion of special shapes of voltage change characteristics into flicker-equivalent voltage jumps. The shape factor is considered in the calculation of the short- and long-term flicker severity, and in the calculation of the flicker impression time.
- ***Rated current, Ir.*** The rated current of the load/s in A .
- ***Starting current ratio, Ia/Ir.*** The ratio of the starting current to the rated current of the load/s, both in A.

3.3.1.3 Generating units

Table ***Generating units*** per row:

- ***Technology.*** The type of connection of the generating units/s installation (3PH, 1PH).
- ***Flk. Coeff. (TypFlicker).*** Selection of *Flicker* type in which to specify flicker data for the generating unit/s. This data should only include maximum values.
- ***Apparent power change, dSa.*** Apparent power change in kVA for generating unit/s, relevant to the assessment of network disturbances.
- ***Angle, phi.*** The angle of generating unit/s in degrees.
- ***Displacement factor, cos(phi).*** The quotient of the active power and the apparent power of the generating unit/s, relative to the fundamental voltage and current.
- ***Active power change, dPa.*** Active power change of generating unit/s in kW , relevant to the assessment of network disturbances.

- ***Reactive power change, dQa.*** Reactive power change of generating unit/s in kvar, relevant to the assessment of network disturbances.
- ***Impedance ratio, alpha (Zn/Zp).*** Ratio of the impedance of the neutral wire (Zn) to the outer conductor (Zp) of generating unit/s. Used for calculation of the relative voltage change at the operating point of the generating unit/s.
- ***Repetition rate, r.*** The repetition rate of generating unit/s in 1/min. *PowerFactory* uses the Pst=1 curve (see [8]) for the calculation of flicker, and therefore supports repetition rates $\leq$ 12000/min.
- ***Shape factor, F.*** The shape factor of generating unit/s. Used for the conversion of special shapes of voltage change characteristics into flicker-equivalent voltage jumps. The shape factor is considered in the calculation of the short- and long-term flicker severity, and in the calculation of the flicker impression time.

3.3.2 Plant load change

- ***Input mode.*** Selection of mode of data input.
- ***Apparent power change, dSa.*** Apparent power change in kVA for devices and installations, relevant to the assessment of network disturbances.
- ***Angle, phi.*** The angle of the installation of the network user in degrees.
- ***Displacement factor, cos(phi).*** The quotient of the active power and the apparent power, relative to the fundamental voltage and current.
- ***Active power change, dPa.*** Active power change in kW for devices and installations, relevant to the assessment of network disturbances.
- ***Reactive power change, dQa.*** Reactive power change in kvar for devices and installations, relevant to the assessment of network disturbances.
- ***Rated voltage, Ur.*** The rated voltage of the motor in V. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Rated current, Ir.*** The rated current of the motor in A. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Starting current ratio, Ia/Ir.*** The ratio of the motor starting current (A) to the motor rated current, both in A. Valid range is [3-8]. Applicable only when *Input Mode* is set to *Motor Data*.
- ***Load technology.*** The type of connection of the user installation (3PH, 1PH PH-PH, 1PH).
- ***Transformer characteristic number.*** The transformer characteristic number.
- ***Repetition rate.*** The repetition rate in 1/min. *PowerFactory* uses the Pst=1 curve (see [8]) for the calculation of flicker, and therefore supports repetition rates $\leq$ 12000/min.
- ***Shape factor.*** The shape factor. Used for the conversion of special shapes of voltage change characteristics into flicker-equivalent voltage jumps. The shape factor is considered in the calculation of the short- and long-term flicker severity, and in the calculation of the flicker impression time.
- ***Summation exponent, alpha.*** The summation exponent α. The summation exponent is considered in the calculation of the resulting long-term flicker severity of the load/s in this row of the table.
- ***Simultaneity factor, g.*** The simultaneity factor for flicker generating load/s in this row of the table.

3.4 Voltage unbalance

3.4.1 Loads and generating units

General quantities used for both loads and generating units.

- ***Neutral wire present.*** Selection of phase configuration for loads and generation units to include a neutral wire.

3.4.2 Loads

- ***Technology.*** The type of connection (3PH, 1PH PH-PH, 1PH) of the load/s in this row of the table.
- ***Phase.*** The phase to which the load/s are connected.

3.4.3 Generating units

- ***Technology.*** The type of connection (3PH or 1PH) of the generating unit/s in this row of the table.
- ***Phase.*** The phase to which the generating unit/s are connected.

3.4.4 Unbalance proportionality factor, s

3.4.4.1 User defined

- ***Value.*** The value of the unbalance proportionality factor is input directly (see [8] for guideline values).

3.4.4.2 Calculated

- ***Grid proportional contribution, Gu.*** Proportional contribution of the grid in %.
- ***Reduction factor, ku.*** Reduction factor, dependant on the grid characteristic. Only relevant for a PCC at LV/MV.
- ***Transformer short-circuit voltage, uk.*** Short-circuit voltage of the step-up transformer in %. Only relevant for a PCC at LV/MV.
- ***Contribution reserved for grid, kN.*** The contribution reserved for the grid related to Gu. Only relevant for a PCC at MV/HV.

3.5 Harmonics

This input data applies to both loads and generating units. It should be noted that Harmonics will only be assessed for generating units whose *Type* on the *Installation* subpage is set to *Generator with converter*.

- ***Input of device data: According to groups.*** Device data used for calculation of the harmonic load. Input according to Groups defined in [8].
- ***Input of device Data: According to total harmonic distortion (THDi).*** Device data used for calculation of the harmonic load. Input according to total harmonic distortion is automatically classified into corresponding Groups (as defined in [8]) and displayed as read-only data in the table.
- ***System capacitance.*** Used in the calculation of the first parallel resonance point. Input must already consider length. If not specified, and option *calculated* is selected, *PowerFactory* will calculate the capacitance considering all lines/cables and purely capacitive shunts connected at the same voltage level as the PCC.
- ***Simultaneity factor, g.*** Used in the calculation of harmonic current emission limits and subsequently, THDiA.
- ***Neutral wire present.*** Selection of phase configuration for loads and generation units to include a neutral wire.

3.6 Commutation notches

- ***Converter rated power.*** Rated power of the converter in kVA. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Number of pulses.*** Number of pulses of converter. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Converter angle.*** Angle of the converter in degrees. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Connection factor.*** Connection factor dependent on the connection type or transformer switching group.
- ***Relative short-circuit voltage of the commutation reactance, ukKom: Typical value based on the curve data.*** The value of ukKom is calculated for typical values of the relative depth of the commutation notch dkom according to [8].
- ***Relative short-circuit voltage of the commutation reactance, ukKom: User-defined.*** The value of ukKom is directly input by the user in %.

4 BDEW, 4th Supplement

The selection of *BDEW, 4th Supplement* as the *Method* in the Connection Request Assessment command carries out a power quality assessment according to [2]. The associated input data for the connection request element is described in this section.

4.1 Power plant definition

- ***Plant PAmax*** The total rated active power of the plant. Calculated as the sum of the rated apparent powers PEmax of the generating unit/s.
- ***Minimum power factor, lambda.*** The minimum power factor at the junction point.

Table ***Generating units*** per row:

- ***Name.*** An arbitrary name used to describe the generating unit/s.
- ***Type.*** The type of generating unit.
- ***Qty.*** The number of generating unit/s.
- ***Rated voltage, Ur.*** The rated voltage of the generating unit/s in this row of the table in kV.
- ***SrE*** The rated apparent power of the generating unit/s.
- ***PEmax*** The rated active power of the generating unit/s. The plant infeed for load flow purposes uses *PEmax* and *Minimum power factor, lambda*.

4.2 Connection topology

- ***Junction point definition*** Automatic search or user-defined selection of the junction point.

4.3 Sudden voltage changes and flicker

Identical units should be defined in a single table row.

Table ***Voltage changes and flicker*** per row:

- ***WTG***: Tick this option if the generating unit/s should be treated as wind turbine generators. The assessment then uses equation (B.2-5) or (B.2-6) from [2].
- ***Flk. Coeff. (TypFlicker)*** Selection of *Flicker* type in which to specify flicker data for the generating unit/s. This data should only include maximum values.
- ***Assess*** Use either ku or ki for assessment.
- ***kimax*** Switched current factor.
- ***Ia*** Starting current of the generating unit/s in kA.
- ***InG*** Nominal current of the generating unit/s in kA.

4.4 Harmonics and interharmonics

- ***Total connectible power, Sgesamt.*** Used in the calculation of harmonic current emission limits.
- ***Apparent connection power, SA.*** Used in the calculation of harmonic current emission limits.
- ***Ripple control frequency.*** The frequency of the ripple control in Hz.

Table per row:

- ***Harmonic currents (TypHmccur).*** Selection of *Harmonic Current Source* type in which to specify harmonics data for the generating unit/s.
- ***Summation*** Method to use for summation of harmonic currents emitted by the generating unit/s.

4.5 Commutation notches

- ***Converter rated power.*** Rated power of the converter. Used in the calculation of the commutation reactance short-circuit voltage.
- ***Number of pulses.*** Number of pulses of converter. Used in the calculation of the commutation reactance short-circuit voltage.

4.6 Short-circuit current

The short-circuit current should be input for the voltage level of the individual units.

Table ***Short-circuit current*** per row:

- ***Short-circuit current.*** The short-circuit current contribution of each generating unit/s in kA.
- ***Rated current.*** The rated current of each generating unit/s in kA.
- ***Factor.*** The factor (specified by BDEW or VDE as appropriate) applying to each generating unit/s.

5 VDE-AR-N 4105 (2011 and 2018)

The selection of *VDE-AR-N 4105* as the *Method* in the Connection Request Assessment command carries out a power quality assessment according to [3] or [6], depending on the *Year* selected in the command. The associated input data for the connection request element is described in this section.

5.1 Power plant definition

5.1.1 Loads

Table ***Loads*** per row:

- ***Name.*** An arbitrary name used to describe the load/s.
- ***Type.*** The type of load.
- ***Qty.*** The number of load/se.
- ***Out of service.*** Out of service status of load/s.
- ***Rated voltage, Ur.*** The rated voltage of the load/s in kV.
- ***SrB*** The rated apparent power of the load/s.
- ***PBmax*** The rated active power of the load/s. The plant infeed for load flow purposes uses *PBmax* and *Minimum power factor, cos(phi)*.

5.1.2 Generating and storage units

Table ***Generating and storage units*** per row:

- ***Name.*** An arbitrary name used to describe the generating unit/s.
- ***Type.*** The type of generating unit.
- ***Qty.*** The number of generating unit/s.
- ***Out of service.*** Out of service status of generating unit/s.
- ***Rated voltage, Ur.*** The rated voltage of the generating unit/s in kV.
- ***SrE*** The rated apparent power of the generating unit/s.
- ***PEmax*** The rated active power of the generating unit/s. The plant infeed for load flow purposes uses *PEmax* and *Minimum power factor, cos(phi*.

5.1.3 Loads and generating units

General quantities used for both loads and generating units.

- ***Minimum power factor, cos(phi).*** The minimum power factor at the junction point.

5.2 Connection topology

- ***Junction point definition*** Automatic search or user-defined selection of the junction point.
- ***Supply transformer or network infeed definition***: Automatic search or user-defined selection of a single supply transformer (or external grid equivalent) used for various assessments. Some assessments (i.e. minimum short-circuit power, commutation notches and harmonics) may not be possible without this being specified.

5.3 Minimum short-circuit power

It should be noted that the method used to calculate the short-circuit power (*SkV*) is the same for both the 2011 and 2018 editions. This page and its inputs are only relevant to the 2018 edition [6]:

- ***Type-1 generation at voltage level:*** The user-defined list of type-1 generators that are to be considered in the calculation of the limit for the minimum short-circuit power at the supply transformer bus. Elements that are valid inclusion are synchronous generators (*ElmSym*), external grids (*ElmXnet*), static generators (*ElmGenstat*) and connection requests (*ElmConreq*).
- ***Type-1 generation at junction point:*** The user-defined list of type-1 generators that are to be considered in the calculation of the limit for the minimum short-circuit power at the junction point. Elements that are valid inclusion are synchronous generators (*ElmSym*), external grids (*ElmXnet*), static generators (*ElmGenstat*) and connection requests (*ElmConreq*).

5.4 Rapid voltage changes and flicker

As for BDEW (please refer to Section 4.3). For the sudden voltage change of the individual units, the following additional inputs are available (only relevant to the 2018 edition [6]:

- ***Switching freq.:*** The frequency of switching operations as defined in [5] (“Tabelle 3”, 2nd column).
- ***Minimum interval:*** The minimum interval between switching operations as defined in [5] (“Tabelle 3”, 3rd column).

For the assessment of flicker, the following additional inputs are available (only relevant to the 2018 edition [6]:

- ***SA_B:*** Used in conjunction with the supply transformer rated power for calculation of the demand factor, kB.
- ***SA_E:*** Used in conjunction with the supply transformer rated power for calculation of the generation factor, kE.
- ***SA_S:*** Used in conjunction with the supply transformer rated power for calculation of the storage factor, kS.
- ***Apparent connection power, SA***: Used in the calculation of flicker limits.
- ***Simultaneity factor:*** Has a default value of ‘1’.

For the assessment of flicker, identical units should be defined in a single table row.

5.5 Unbalance

- ***Technology.*** The type of connection (3PH or 1PH) of the generating unit/s in this row of the table.
- ***Phase.*** The phase to which the generating unit/s are connected.

5.6 Harmonics and interharmonics

As for BDEW (please refer to Section 4.4). The assessment is undertaken assuming that $I_A > 75$ A. The following additional inputs are required for assessment according to the 2018 edition [6]:

- ***Resonance factor for harmonics:*** Used for the calculation of harmonic current limits. Has a default value of '1.5' or can be user-defined. Please note that this factor will be used for harmonic orders h$\leq$19; and set to kv=1 for harmonic orders h$>$19.
- ***Resonance factor for interharmonics:*** Used for the calculation of interharmonic current limits as defined in Section 5.4.4.4 of [5].
- ***Resonance factor for supraharmonics:*** Used for the calculation of supraharmonic current limits as defined in Section 5.4.4.4 of [5].
- ***SA_B:*** Used in conjunction with the supply transformer rated power for calculation of the demand factor, kB.
- ***SA_E:*** Used in conjunction with the supply transformer rated power for calculation of the generation factor, kE.
- ***SA_S:*** Used in conjunction with the supply transformer rated power for calculation of the storage factor, kS.
- ***Neutral wire:*** Considered when calculating supraharmonic limits (application of scaling factor defined in Section 5.4.4.4 of [5]).

5.7 Commutation notches

- ***Converter rated power.*** Rated power of the converter.
- ***Number of pulses.*** Number of pulses of converter.
- ***Worst-case converter angle.*** Worst-case angle of the converter in degrees.
- ***Relative short-circuit voltage of the commutation reactance.*** Relative short-circuit voltage of the commutation reactance ukKom in %.

5.8 Short-circuit current

As for BDEW (please refer to Section 4.6).

6 VDE-AR-N 4110 (2018)

The selection of *VDE-AR-N 4110 (2018)* as the *Method* in the Connection Request Assessment command carries out a power quality assessment according to [7]. The associated input data for the connection request element is described in this section.

6.1 Installation definition

6.1.1 Loads

Table ***Loads*** per row:

- ***Name.*** An arbitrary name used to describe the load/s.
- ***Type.*** The type of load.
- ***Qty.*** The number of load/s.
- ***Out of service.*** Out of service status of load/s.
- ***Rated voltage, Ur.*** The rated voltage of the load/s in kV.
- ***SrB*** The rated apparent power of the load/s.
- ***PBmax*** The rated active power of the load/s.
- ***Minimum power factor, cos(phiB).*** The minimum power factor of the load/s.

6.1.2 Generating and storage units

Table ***Generating units*** per row:

- ***Name.*** An arbitrary name used to describe the generating unit/s.
- ***Type.*** The type of generating unit.
- ***Qty.*** The number of generating unit/s.
- ***Out of service.*** Out of service status of generating unit/s.
- ***Rated voltage, Ur.*** The rated voltage of the generating unit/s in kV.
- ***SrE*** The rated apparent power of the generating unit/s.
- ***PEmax*** The rated active power of the generating unit/s.
- ***Minimum power factor, cos(phiE).*** The minimum power factor of the generating unit/s.

6.1.3 Loads and generating units

General quantities used for both loads and generating units.

- ***Minimum power factor, lambda.*** The minimum power factor at the junction point.

6.2 Connection topology

- ***Junction point definition*** Automatic search or user-defined selection of the junction point.
- ***Supply transformer or network infeed definition***: Automatic search or user-defined selection of a single supply transformer (or external grid equivalent) used for various assessments. Some assessments (i.e. minimum short-circuit power, commutation notches and harmonics) may not be possible without this being specified.

6.3 Minimum short-circuit power

The following inputs are required:

- ***Type-1 generation at voltage level:*** The user-defined list of type-1 generators that are to be considered in the calculation of the limit for the minimum short-circuit power at the supply transformer bus. Elements that are valid inclusion are synchronous generators (*ElmSym*), external grids (*ElmXnet*), static generators (*ElmGenstat*) and connection requests (*ElmConreq*).
- ***Type-1 generation at junction point:*** The user-defined list of type-1 generators that are to be considered in the calculation of the limit for the minimum short-circuit power at the junction point. Elements that are valid inclusion are synchronous generators (*ElmSym*), external grids (*ElmXnet*), static generators (*ElmGenstat*) and connection requests (*ElmConreq*).

For the assessment of the minimum short-circuit power at the supply transformer bus, all downstream (MV and LV) generation (and load) is outaged and a 3-phase short-circuit (minimum short-circuit currents) according to DIN EN 60909-0 (VDE 0102) is calculated. The resistances of lines and cables are considered at 20 degrees Celsius.

For the assessment of the minimum short-circuit power at the junction point, all MV (and LV) generation (and load) is outaged and a 3-phase short-circuit (minimum short-circuit currents) according to DIN EN 60909-0 (VDE 0102) is calculated. The resistances of lines and cables are considered at 20 degrees Celsius.

6.4 Sudden voltage changes

As for BDEW (please refer to Section 4.3). For loads, the inputs are as defined for D-A-CH-CZ (please refer to Section 2.3). For the sudden voltage change of the individual units, the following additional inputs are available:

- ***Switching freq.:*** The frequency of switching operations as defined in [7] (“Tabelle 2”, 2nd column).
- ***Minimum interval:*** The minimum interval between switching operations as defined in [7] (“Tabelle 2”, 3rd column).

It should be noted that the sudden voltage change of the individual units is calculated directly at the junction point set in the connection request element.

6.5 Flicker

As for BDEW (please refer to Section 4.3). For the calculation of the short-term flicker (Pst) for loads, inputs are as defined for D-A-CH-CZ. Please refer to Section 2.3. The following additional inputs are required:

- ***SA_B:*** Used in conjunction with the supply transformer rated power for calculation of the demand factor, kB.
- ***SA_E:*** Used in conjunction with the supply transformer rated power for calculation of the generation factor, kE.
- ***SA_S:*** Used in conjunction with the supply transformer rated power for calculation of the storage factor, kS.
- ***Simultaneity factor:*** Has a default value of '1'.

For the assessment of flicker, identical units should be defined in a single table row.

6.6 Harmonics, interharmonics and supraharmonics

As for BDEW (please refer to Section 4.4). Please refer to definitions of duplicate inputs on the Flicker subpage in Section 6.5. The following additional inputs are required:

- ***Apparent connection power, SA.*** Used in the calculation of harmonic current emission limits. For VDE4110, this is automatically calculated and considers the in-service status of each unit in the plant.
- ***Resonance factor for harmonics:*** Used for the calculation of harmonic current limits as defined in Section 5.4.4 of [7]. Has a default value of '1.5' or can be user-defined. Please note that this factor will be used for harmonic orders $h \leq 19$; and set to kv=1 for harmonic orders $h > 19$.
- ***Resonance factor for interharmonics:*** Used for the calculation of interharmonic current limits as defined in Section 5.4.4 of [7].
- ***Resonance factor for supraharmonics:*** Used for the calculation of supraharmonic current limits as defined in Section 5.4.4 of [7].

6.7 Commutation notches

- ***Converter rated power.*** Rated power of the converter.
- ***Number of pulses.*** Number of pulses of converter.
- ***Worst-case converter angle.*** Worst-case angle of the converter in degrees.
- ***Relative short-circuit voltage of the commutation reactance.*** Relative short-circuit voltage of the commutation reactance ukKom in p.u..

6.8 Short-circuit current

Inputs are as described for BDEW (please refer to Section 4.6), however it must be noted that **the short-circuit current should be input for the voltage level of the point of connection (PoC) of the plant**. The short-circuit contribution is considered for generating/storage units.

7 References

[1] Technical Rules for the Assessment of Network Disturbances, 2nd Edition, 2007.

[2] Regelungen und Uebergangsfristen fuer bestimmte Anforderungen in Ergaenzung zur technischen Richtlinie - Erzeugungsanlagen am Mittelspannungsnetz - Richtlinie fuer Anschluss und Parallelbetrieb von Erzeugungsanlagen am Niederspannungsnetz, 2008.

[3] VDE-AR-N 4105: Technische Mindestanforderungen fuer Anschluss und Parallelbetrieb von Erzeugungsanlagen am Niederspannungsnetz (Generators connected to the low-voltage distribution network - Technical requirements for the connection to and parallel operation with low-voltage distribution networks), 2011.

[4] Technical Rules for the Assessment of Network Disturbances, 2nd Edition - HV Extension (Ergaenzungsdokument zur Beurteilung von Anlagen fuer den Anschluss an Hochspannungsverteilernetze), 2012.

[5] VDE-AR-N 4100: Technische Regeln fuer den Anschluss von Kundenanlagen an das Niederspannungsnetz und deren Betrieb (TAR Niederspannung) (Technical rules for the connection and operation of customer installations to the low voltage network (tar low voltage)), 2018.

[6] VDE-AR-N 4105: Technische Mindestanforderungen fuer Anschluss und Parallelbetrieb von Erzeugungsanlagen am Niederspannungsnetz (Generators connected to the low-voltage distribution network - Technical requirements for the connection to and parallel operation with low-voltage distribution networks), 2018.

[7] VDE-AR-N 4110: Technische Regeln fr den Anschluss von Kundenanlagen an das Mittelspannungsnetz und deren Betrieb (TAR Mittelspannung) (Technical requirements for the connection and operation of customer installations to the medium voltage network (TAR medium voltage)), 2018.

[8] Technical Rules for the Assessment of Network Disturbances, 3rd Edition, 2021.